

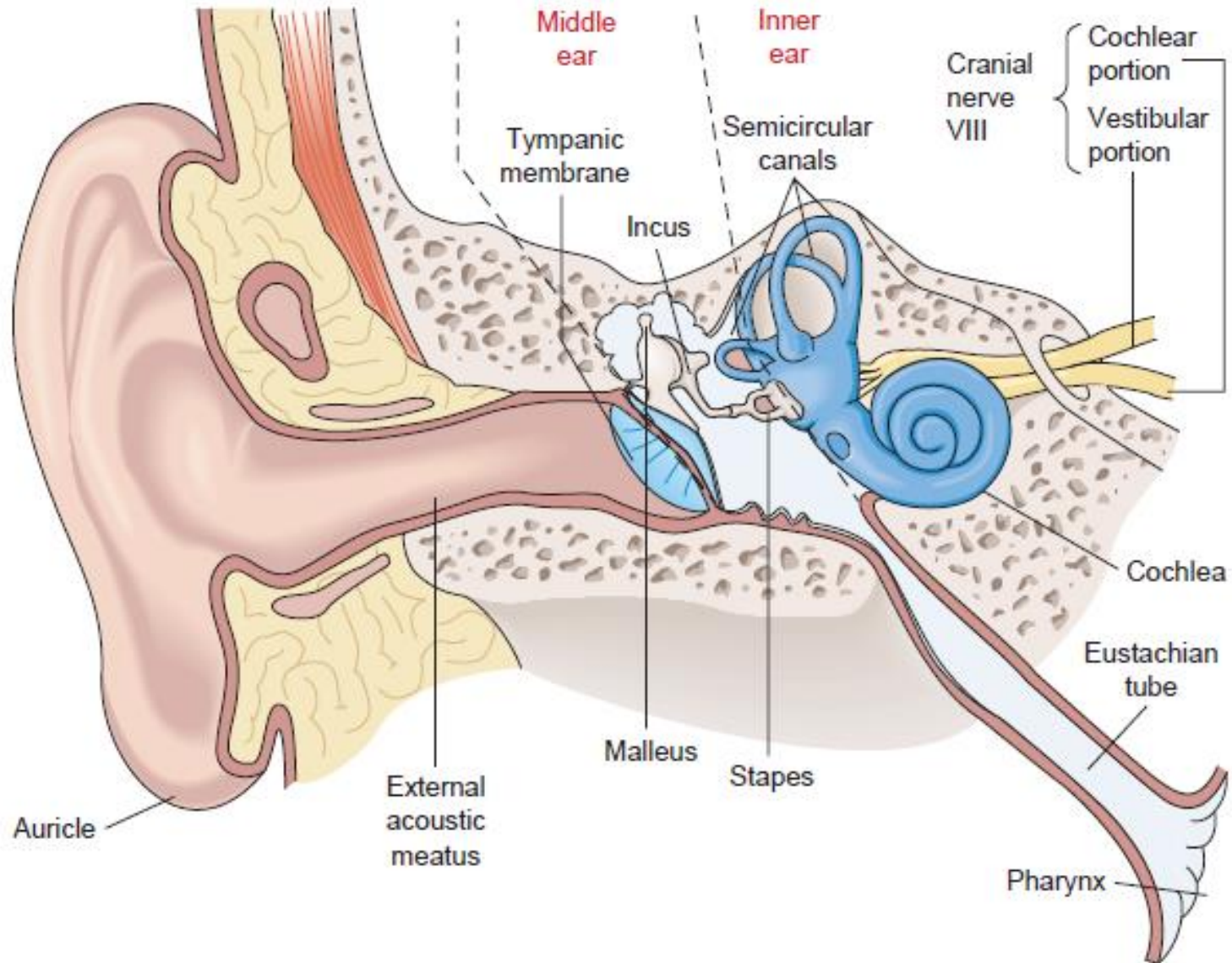
ENT

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Anatomic & Physiologic Overview of Ear

- The ears are a pair of complex sensory organs located in the middle of both sides of the head at approximately eye level
- The external ear, housed in the temporal bone, includes the auricle (i.e., pinna) and the external auditory canal.
- The external ear is separated from the middle ear by a disklike structure called the tympanic membrane (i.e., eardrum).

Ear anatomy



Auricle

- The auricle, attached to the side of the head by skin, is composed mainly of cartilage, except for the fat and subcutaneous tissue in the earlobe.
- The function of auricle is to collect the sound waves and directs vibrations into the external auditory canal.

External Auditory Canal

- Is approximately 2.5 cm long.
- A thin layer of skin containing fine hairs, sebaceous glands, and ceruminous glands lines the ear canal.
- These glands produce cerumen, or earwax, which has certain antimicrobial properties and is thought to serve a protective function.
- The external auditory canal ends at the tympanic membrane.
- The ear's self-cleaning mechanism moves old skin cells and cerumen to the outer part of the ear.

Anatomy of the middle ear

- The middle ear, an air-filled cavity, includes the tympanic membrane laterally and the otic capsule medially.
- The middle ear is connected to nasopharynx by eustachian tube
- Normally, the tube is closed, but it opens during yawning or swallowing.
- The tube serves as a drainage channel for normal and abnormal secretions of the middle ear and equalizes pressure in the middle ear with that of the atmosphere.

Tympanic Membrane

- The tympanic membrane (i.e., eardrum), about 1 cm in diameter and very thin, is normally pearly gray and translucent.
- The tympanic membrane consists of three layers of tissue: an outer layer, continuous with the skin of the ear canal; a fibrous middle layer; and an inner mucosal layer, continuous with the lining of the middle ear cavity.
- The tympanic membrane protects the middle ear and conducts sound vibrations from the external canal to the ossicles.

Ossicles

- The middle ear contains the three smallest bones (i.e., ossicles) of the body: - malleus, - incus, and - stapes.
- It has vibratory, resonance function and modify the external stimulus.
- The ossicles, which are held in place by joints, muscles, and ligaments, assist in the transmission of sound.

Anatomy of the inner ear

- The inner ear is housed deep within the temporal bone.
- The organs for hearing (i.e., cochlea) and balance (i.e., semicircular canals), as well as cranial nerves VII (i.e., facial nerve) and VIII (i.e., vestibulocochlear nerve), are all part of this complex anatomy.
- The cochlea and semicircular canals are housed in the bony labyrinth.
- The bony labyrinth surrounds and protects the membranous labyrinth, which is bathed in a fluid called perilymph.

Membranous Labyrinth

- Is composed of the utricle, the saccule, the cochlear duct, the semicircular canals, and the organ of Corti.
- Contains a fluid called endolymph.
- The three semicircular canals—posterior, superior, and lateral, which lie at 90-degree angles to one another—contain sensory receptor organs, arranged to detect rotational movement.
- These receptor end organs are stimulated by changes in the rate or direction of an individual's movement.

Organ of Corti

- The organ of Corti is located in the cochlea, a snail-shaped, bony tube about 3.5 cm long with two and one-half spiral turns.
- The organ of Corti, also called the end organ for hearing, transforms mechanical energy into neural activity and separates sounds into different frequencies.

Function of the ears

- Hearing:- Hearing is conducted over two pathways: air and bone.
- Sounds transmitted by air conduction travel over the air- filled external and middle ear through vibration of the tympanic membrane and ossicles.
- Sounds transmitted by bone conduction travel directly through bone to the inner ear, bypassing the tympanic membrane and ossicles.
- Normally, air conduction is the more efficient pathway. (AC>BC)

Sound conduction and transmission

- Sound enters the ear through the external auditory canal causes the tympanic membrane to vibrate.
- These vibrations transmit sound through the action of the ossicles to the oval window as mechanical energy.
- This mechanical energy is then transmitted through the inner ear fluids to the cochlea, stimulating the hair cells, and is subsequently converted to electrical energy.

Sound conduction and transmission ...

- The electrical energy travels through the vestibulocochlear nerve to the central nervous system, where it is analyzed and interpreted in its final form as sound.

Assessment

- HEALTH HISTORY:- It includes all the components that are applied in other body system.
 - Date of History.
 - Identification.
 - Chief complaint.
 - History of present illness.
 - History of past illness.
 - Current health status (Current medication, addictive drugs and allergies).
 - Family history.
 - Psychosocial and personal history.

Physical Examination

- The external ear is examined by; Inspection of external ear :- for the presence of scar, lesion, symmetry, attachment, any abnormal discharge, color e.t.c.
- Tympanic membrane is inspected with an otoscope.
- Inspection of the middle ear with middle ear endoscopy.
- Direct palpation:- for tenderness, presence of malignancy, free movement, circulation, e.t.c.

P/E of ear

❖ On palpation

- They should feel firm, and movement should produce no pain.
- Palpating the mastoid process should be painless.
- Abnormal- pain with movement occurs with **otitis externa**
- pain at the mastoid process may indicate **mastoiditis**

Otoscopic Exam



Otoscopic examination

- Used to examine the external auditory canal and tympanic membrane
- Before inserting the otoscope it is important to straighten the external auditory canal by manipulation;
- Grasp the auricle firmly but gently and pull it **upward**, backward, and slightly away from the head in adult.

Otoscopic examination...

- Grasp the auricle firmly but gently and pull it **down ward**, backward, and slightly away from the head in Children.
- Once it is in place, you may need to rotate the otoscope slightly to visualize the entire drum; do this gently
- Proper otoscopic examination of the external auditory canal and tympanic membrane requires that the canal be free of large amounts of cerumen.

Otoscopic examination...

- In the external cannal note any redness, swelling, lesions, foreign bodies, or discharge
- Purulent pus discharge may indicate otitis media if the drum has ruptured
- The healthy tympanic membrane is pearly gray and is positioned obliquely at the base of the canal.
- The normal ear drum is shinny and translucent, with a gray color

Evaluation of gross auditory acuity

- A general estimation of hearing can be made by assessing the patient's by;
 - Whisper test
 - Weber
 - Rinne tests

Whisper Test

- To exclude one ear from the testing, the examiner covers the untested ear with the palm of the hand.
- Then the examiner whispers softly from a distance of 1 or 2 feet from the unoccluded ear and out of the patient's sight.
- The patient with normal acuity can correctly repeat what was whispered.

Weber Test (Lateralization Test)

- The Weber test uses bone conduction to test lateralization of sound.
- A tuning fork (ideally, 512 Hz), set in motion by grasping it firmly by its stem and tapping it on the examiner's knee or hand, is placed on the patient's head or forehead.
- A person with normal hearing will hear the sound equally in both ears or describe the sound as centered in the middle of the head.

Weber Test...

- In cases of **conductive hearing loss**, such as from otosclerosis or otitis media, the sound is heard better in the affected ear.
- In cases of **sensorineural hearing loss**, resulting from damage to the cochlear or vestibulocochlear nerve, the sound lateralizes to the better-hearing ear.
- The Weber test is useful for detecting unilateral hearing loss.

Weber test



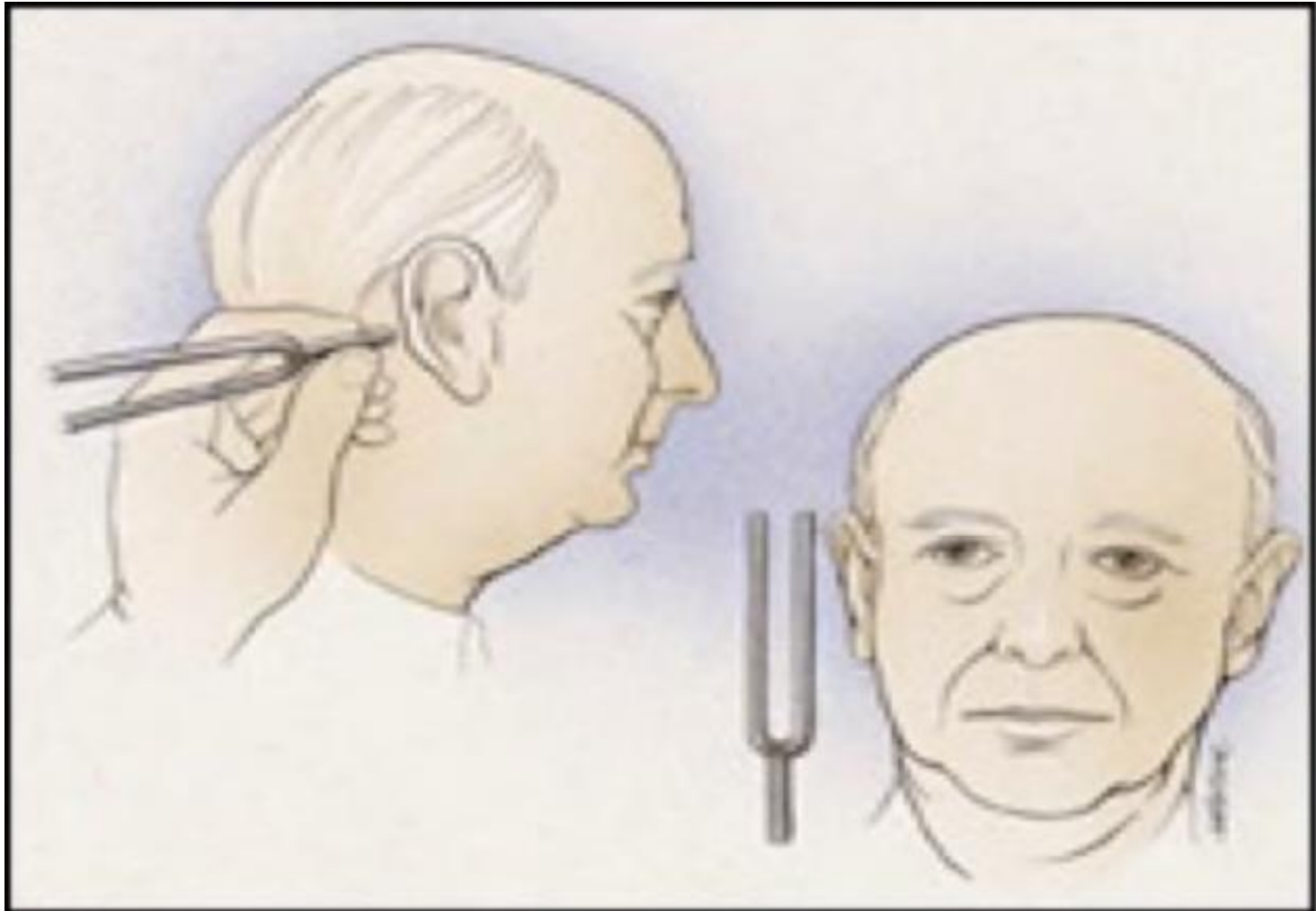
Rinne Test

- In the Rinne test the examiner shifts the stem of a vibrating tuning fork between two positions:
 - 2 inches from the opening of the ear canal (i.e., for air conduction) and against the mastoid bone (i.e., for bone conduction).
- Normally, sound heard by air conduction is audible longer than sound heard by bone conduction.
- The Rinne test is useful for distinguishing between conductive and sensorineural hearing losses.

Rinne Test

- With a conductive hearing loss, bone-conducted sound is heard as long as or longer than air-conducted sound, whereas with a sensorineural hearing loss, air-conducted sound is audible longer than bone conducted sound.

Rinne's test



Diagnostic Evaluation

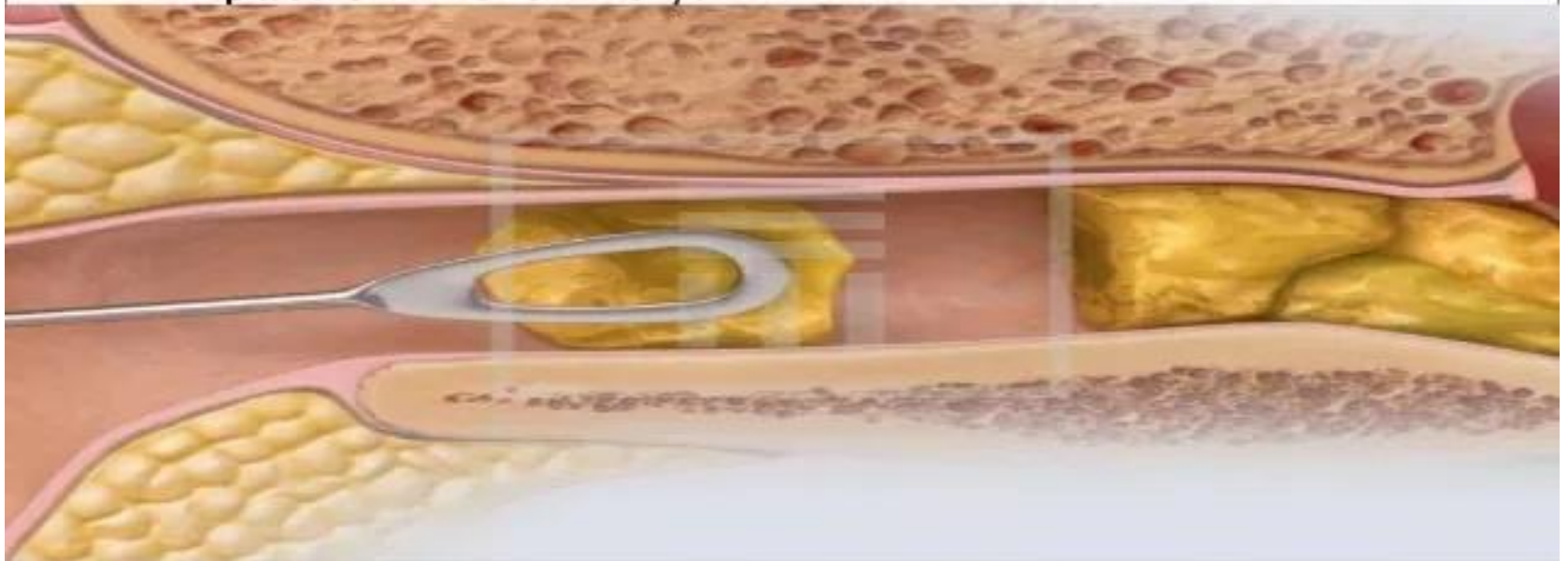
1. Audiometry { test to check the ability of hear sounds,}
2. Tympanogram {tympanometry is an examination used to test the condition of the middle ear and tympanic membrane}
3. Auditory brain stem response {ABR- is an extracted from ongoing electrical activity in the brain and recorded via electrodes placed on the scalp.}
4. Electronystagmography{ ENG- used to evaluate patients with dizziness, vertigo, or balance dysfunction or to record involuntary movements of the eyes caused by a condition known as nystagmus.}
5. Platform posturography{ TOB- TEST OF BLANCE}
6. Sinusoidal harmonic acceleration{ TO CHECK THE VESTIBULAR FUNCTION}
7. Middle ear endoscopy

External Ear disorder

1. CERUMEN IMPACTION

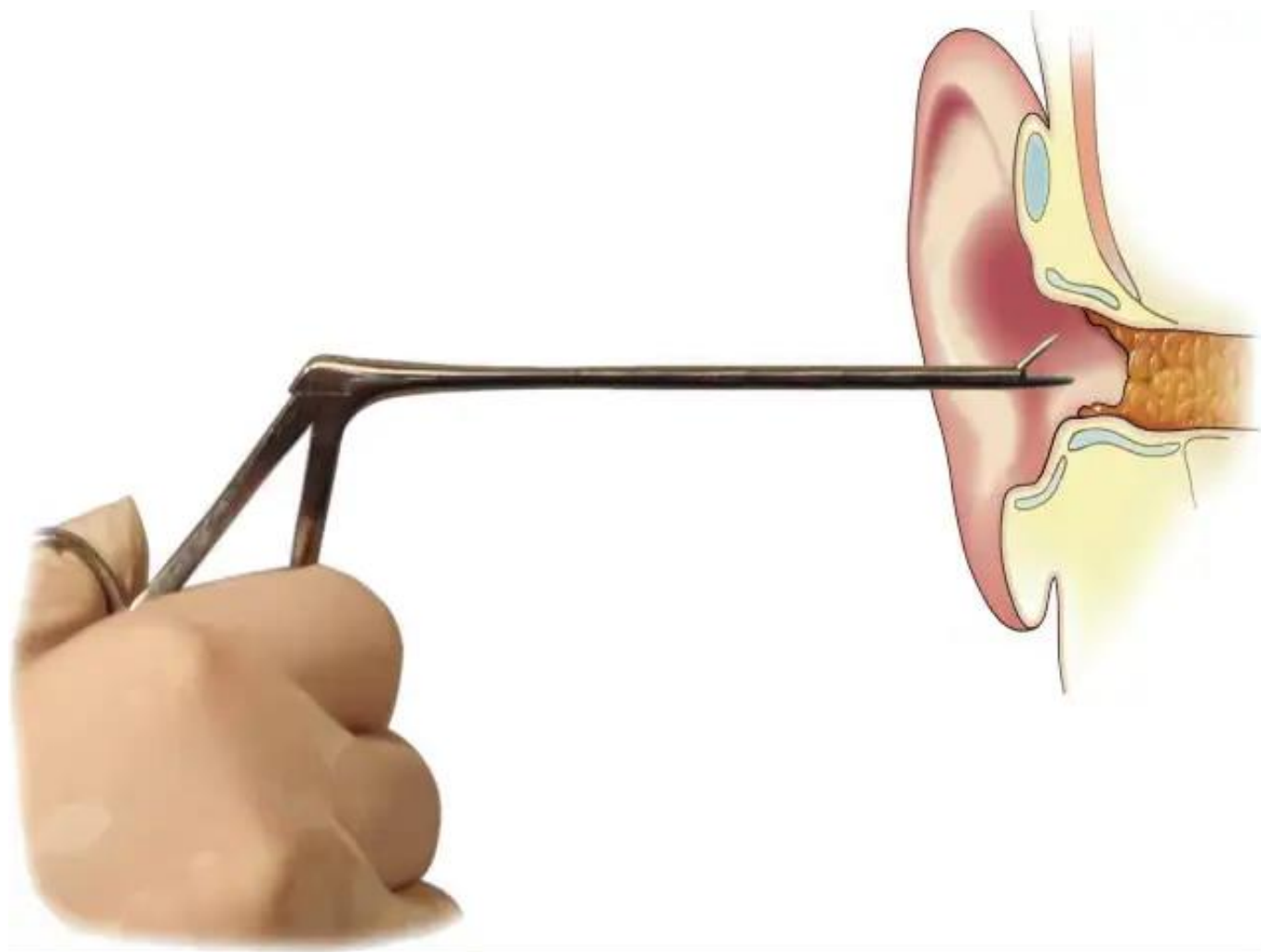
- Cerumen normally accumulates in the external canal in various amounts and colors.
- Although wax does not usually need to be removed, impaction occasionally occurs, causing otalgia, a sensation of fullness or pain in the ear, with or without a hearing loss.
- Accumulation of cerumen is especially significant in the geriatric population as a cause of hearing deficit.

- - Accumulated cerumen (earwax) may become impacted due to use of cotton swabs to clean ears and may be a problem for some people.
 - Cerumen becomes drier in elderly people, making impaction more likely.



Cerumen impaction clinical manifestation

- Hearing loss
- Stuffed or full feeling to ear
- Pain if cerumen touch TM



Management

❖ Cerumen can be removed by

- ✓ Irrigation: (Unless the patient has a perforated eardrum or an inflamed external ear (i.e., otitis externa), particularly if it is not tightly packed in the external auditory canal).
- ✓ Suction: Using any softening solution two or three times a day for several days is generally sufficient.
- Instilling a few drops of warmed glycerin, mineral oil, or half strength hydrogen peroxide into the ear canal for 30 minutes can soften cerumen before its removal.

Management....

✓ Instrumentation.

- If the cerumen cannot be dislodge by these methods, instruments, such as a cerumen curette, aural suction, and a binocular microscope for magnification, can be used.
- Direct visual, mechanical removal can be performed on a cooperative patient by a trained health care provider.)
- To prevent injury, the lowest effective pressure should be used.
- Ceruminolytic agents, such as peroxide in glyceryl (Debrox), are available; however, these compounds may cause an allergic dermatitis reaction.

Foreign bodies

- Some objects are inserted intentionally into the ear by adults who may have been trying to clean the external canal or relieve itching or by children who introduce the objects.
- Other objects, such as insects, peas, beans, pebbles (Sand/stone), toys, and beads/droplet, may enter or be introduced into the ear canal.
- In either case, the effects may range from no symptoms to profound pain and decreased hearing.
- C/M –No symptoms, - Swelling, - Profound pain, - Decreased hearing,

Management

- The three standard methods for removing foreign bodies are the same as those for removing cerumen:
 - Irrigation: Foreign vegetable bodies and insects tend to swell; thus, irrigation is contraindicated.,
 - Suction, and
 - Instrumentation.
- Usually, an insect can be dislodged by instilling mineral oil, which will kill the insect and allow it to be removed.
- In difficult cases, the foreign body may have to be extracted in the operating room with the patient under general anesthesia.

External otitis (otitis externa)

- It is an inflammation of the external auditory canal.
- Usually caused by acute infection.
- Causes
 - Water in the ear canal (i.e., swimmer's ear).
 - Trauma to the skin of the ear canal.
 - Systemic conditions (such as vitamin deficiency (Vit.A) and endocrine disorders).

otitis externa.....

- Causes
- Bacterial infections (most common are *Staphylococcus aureus* and *Pseudomonas* species).
- Fungal infection (most common is *Aspergillus*).
- Dermatitis (such as psoriasis, eczema, or seborrheic dermatitis).
- Allergic reactions to hair spray, hair dye, and permanent wave lotions can cause dermatitis, which clears when the offending agent is removed.

otitis externa

❖ Clinical Manifestations

- Pain,
- Discharge (yellow or green and foul smelling),
- Aural tenderness (usually not present in middle ear infections),
- Fever,
- Cellulitis,

Otitis externa

❖ Clinical Manifestations

- Lymphadenopathy,
- Pruritus,
- Hearing loss,
- Feeling of fullness,
- Erythematous and edematous (otoscopic examination),
- In fungal infections, the hair like black spores may even be visible.

Medical Management

- The principles of therapy are aimed at; relieving the discomfort, reducing the swelling of the ear canal, and eradicating the infection.
- Patients may require analgesics for the first 48 to 92 hours.
- Antibiotic ear drops
- Antifungal- clotrimazole ear drop 1% 2-3 times/d at least for 14 days.
- Avoid swimming & do not allow water to enter the ear.

Nursing Management

- Nurses need to teach patients;
 - Not to clean the external auditory canal with cotton-tipped applicators,
 - To avoid swimming, and
 - Not to allow water to enter the ear when shampooing or showering.
- A cotton ball can be covered in a water-insoluble gel such as petroleum jelly and placed in the ear as a barrier to water contamination.
- Infection can be prevented by using antiseptic otic preparations after swimming (eg, Swim Ear, Ear Dry).

Middle Ear disorder

❖ Tympanic membrane perforation

- Causes
 - Infection.
 - Trauma (skull fracture, explosive injury, or a severe blow to the ear).
 - Foreign objects (eg, cotton-tipped applicators, match pins, keys) that have been pushed too far into the external auditory canal.



Normal eardrum



Ruptured eardrum

Medical Management

- Most tympanic membrane perforations heal spontaneously within weeks after rupture.
- In the case of a head injury or temporal bone fracture, a patient is observed for evidence of cerebrospinal fluid leakage, otorrhea or rhinorrhea (a clear, watery drainage from the ear or nose), respectively.
- While healing, the ear must be protected from water.

Surgical management

- Tympanoplasty (i.e., surgical repair of the tympanic membrane).
- Surgery is usually successful in closing the perforation permanently and improving hearing

Acute otitis media

- It is an acute infection of the middle ear, usually lasting less than 6 weeks.
- Acute otitis media is an inflammation and infection of the middle ear caused by the entrance of pathogenic organisms, with rapid onset of signs and symptoms.
- It is a major problem in children but may occur at any age.

Causes

- Primarily Streptococcus pneumoniae, Haemophilus influenzae, and Moraxella catarrhalis.
- Inflammation of surrounding structures (eg, sinusitis, adenoid hypertrophy).
- Allergic reactions (eg, allergic rhinitis).
- It is usually present in the middle ear, resulting in a conductive hearing loss.

Clinical Manifestations

- Otalgia : ear pain (unilateral in adults) may awaken patient at night.
 - Pain relieved after tympanic perforation.
- Drainage from the ear (purulent exudate).
- Fever, headache
- Hearing loss (conductive hearing loss).
- The patient reports no pain with movement of the auricle.
- The tympanic membrane is erythematous and often bulging.

Diagnostic Evaluation

- Pneumatic otoscopy shows a tympanic membrane that is full, bulging, and opaque with impaired mobility (or retracted with impaired mobility).
- Cultures of discharge through ruptured tympanic membrane may suggest causative organism

Medical Management

- Antibiotics:-
 - Co-trimoxazole, 4mg/kg trimethoprin 20mg/kg sulphomethaxazole twice a day for 05 days.
 - Amoxicillin, 20-40mg/kg/day divided into 3 doses po/for 5 days
- Clean the external auditory canal
- Cover with cotton

Surgical management

- An incision in the tympanic membrane is known as myringotomy or tympanotomy.
- The incision heals within 24 to 72 hours.
- Indication;
 - For analysis of drainage and if pain persists
- If episodes of acute otitis media recur and there is no contraindication, a ventilating, or pressure-equalizing tube may be inserted.
- The ventilating tube, which temporarily takes the place of the eustachian tube in equalizing pressure, is retained for 6 to 18 months.
- Ventilating tubes are more commonly used to treat recurrent episodes of acute otitis media in children than in adults.

Complications

- Chronic OM
- Mastoiditis
- Meningitis
- Brain abscess

Comparison of AOE and AOM

Feature	Acute otitis externa	Acute otitis media
Otorrhea	May or may not be present	Present if tympanic membrane perforates , drainage profuse
Otalgia	Persistent, may awaken patient at night	Relieved if tympanic membrane rupture
Aural tenderness	Present on palpation of auricle	Usually absent
Systemic symptoms	Absent	Fever, upper respiratory infection, rhinitis
Edema external auditory canal	Present	Absent
Tympanic membrane	May appear normal	Erythema, may be profound
Hearing loss	Conductive type	Conductive type

Serous otitis media

- Serous otitis media (i.e., middle ear effusion) implies fluid, without evidence of active infection, in the middle ear.
- Causes
- Children:-
 - Eustachian tube obstruction (-ve pressure in the middle ear)
- Adults:-
 - Eustachian tube dysfunction (concurrent URTI or allergy)
 - Radiation therapy.
 - Barotrauma(results from sudden pressure changes in the middle ear caused by changes in barometric pressure, as in scuba diving or airplane descent.
 - Carcinoma (eg, nasopharyngeal cancer)

Clinical Manifestations

- Hearing loss (conductive hearing loss),
- Fullness in ear,
- Sensation of congestion,
- Popping and crackling noises,
- Dull tympanic membrane

Diagnosis

- Otoscope-dull TM, and air bubble shown in the middle ear.
- Audiogram- to exclude conductive hearing loss.
- Management
 - Myringotomy.(To relieve pressure or drain fluid by incision)
 - Tube may be placed to keep the middle ear ventilated.
 - Corticosteroids.
 - Valsalva maneuver (do cautiously)

Chronic otitis media

- Is the result of repeated episodes of acute otitis media causing irreversible tissue pathology and persistent perforation of the tympanic membrane.
- Chronic infections of the middle ear damage the tympanic membrane, destroy the ossicles, and involve the mastoid.
- The most common organisms are group A beta-hemolytic streptococci, *S. pneumoniae*, and *H. influenzae*.
- Other organisms may be present, such as *Pseudomonas*, *Proteus*, and *Bacteroides* species

Clinical Manifestations

- Presence of a persistent or intermittent, foul-smelling otorrhea .
- Pain is not usually experienced, except in cases of acute mastoiditis.
- Otoscopic exam;
 - Perforated tympanic membrane.
 - Cholesteatoma (an ingrowth of the skin of the external layer of the eardrum into the middle ear).
 - Audiometric tests often show a conductive or mixed hearing loss

Diagnostic Evaluation

- Air conductive hearing loss is present through audiometric tests.
- X-rays may show mastoid pathology, for example, cholesteatoma or haziness of mastoid cells.
- Culture of exudate from middle ear (through ruptured tympanic membrane or at time of surgery)

Medical Management

- Suctioning of the ear.
- Instillation of antibiotic drops or application of antibiotic powder.
- Systemic antibiotics are usually not prescribed except in cases of acute infection.
- Dry the ear by wicking

Surgical management

- Tympanoplasty (most common surgical procedure).
- There are five types of tympanoplasties.
 - Type I (myringoplasty)-closing the perforated TM, and it is the simplest.
 - Types II through V-more extensive.
 - Ossiculoplasty (surgical reconstruction of the middle ear bones-ossicles).
 - Mastoidectomy (The objectives of mastoid surgery are to remove the cholesteatoma, gain access to diseased structures, and create a dry and healthy ear)

Complications

- Facial nerve palsy.
- Chronic mastoiditis.
- Meningitis.
- Brain abscess.

Inner Ear disorder

- ❖ Common complaint that individual with IED are;
 - Dizziness (any altered sensation of orientation in space).
 - Vertigo (the misperception or illusion of motion of the person or the surroundings). Most people with vertigo describe a spinning sensation or say they feel as though objects are moving around them.
 - Ataxia (failure of muscular coordination due to vestibular system).
 - Nystagmus (an involuntary rhythmic movement of the eyes). can be horizontal, vertical, or rotary.

Motion sickness

- Motion sickness is a disturbance of equilibrium caused by constant motion (aboard a ship, while riding on a merry-go-round or swing, or in the back seat of a car) that over stimulate the vestibular system.
- Clinical Manifestations
- Sweating, Pallor, vertigo, Nausea, and Vomiting.
- These manifestations may persist for several hours after the stimulation stops.

Cont....

- Management
- Over-the-counter antihistamines.
 - dimenhydrinate (Dramamine) or
 - meclizine hydrochloride (Bonine),
 - Anticholinergic medications (scopolamine patch, promethazine, e.t.c.).
 - Avoid potentially hazardous activities (operating heavy machinery, driving a car).

Ménière's disease

- Ménière's disease is an abnormal inner ear fluid balance caused by a malabsorption in the endolymphatic sac.
- Ménière's disease (endolymphatic hydrops) is a chronic disease that involves the inner ear and causes a triad of symptoms
 - Vertigo, hearing loss, and tinnitus.
- Ménière's disease appears to be equally common in both genders.

Clinical Manifestations

- Sudden attacks occur, in which patient feels that the room is spinning (vertigo); may last 10 minutes to several hours.
- Dizziness, tinnitus, and reduced hearing occur on involved side.
- Headache, nausea, vomiting, and incoordination are present.
- Sudden motion of the head may precipitate vomiting.
- Irritability; other personality changes.
- After multiple attacks, tinnitus and impaired hearing may be continuous

Cont....

Assessment and Diagnostic Findings

- Hx.
- P/E.
- Audiogram.
- Electronystagmogram.

Cont....

Pharmacologic therapy

- Antihistamines (meclizine (Antivert)).
- Tranquilizers (diazepam (Valium)).
- Antiemetics (promethazine (Phenergan))
suppositories.
- Diuretic therapy (eg, hydrochlorothiazide).
- Vasodilators (nicotinic acid, papaverine
hydrochloride (Pavabid), and methantheline
bromide (Banthine))

Cont....

❖ Surgical management

- Destructive surgery:
 - Labyrinthectomy : recommended if the patient experiences progressive hearing loss and severe vertigo attacks so normal tasks cannot be performed; results in total deafness of affected ear.

Labyrinthitis

- Labyrinthitis is an inflammation of the inner ear vestibular labyrinth system.
- It may be due to a viral or bacterial infection.

Causes

- Bacterial (complication of otitis media).
- Viral in origin (mumps, rubella, rubeola, and influenza)
- Viral illnesses of the upper respiratory tract.
- Herpetiform disorders of the facial and acoustic nerves (i.e., Ramsay Hunt syndrome).

Cont....

❖ Clinical Manifestations

- Sudden onset of incapacitating vertigo,
- Nausea,
- Vomiting,
- Various degrees of hearing loss, and
- Tinnitus.

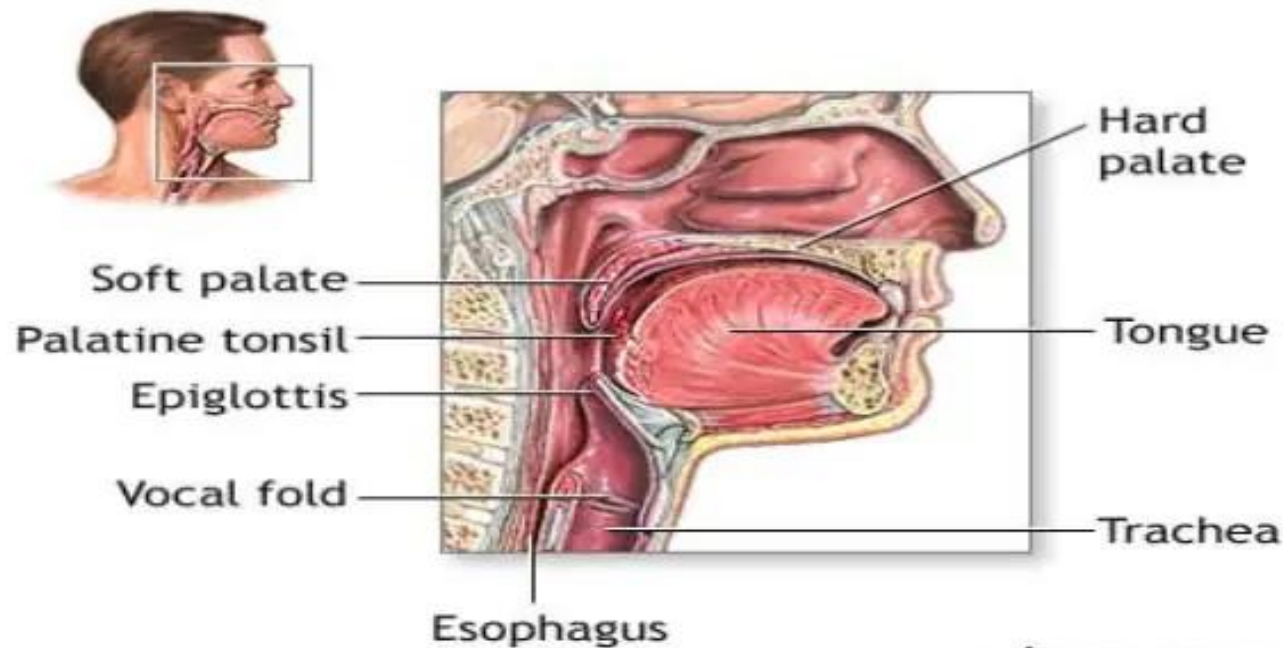
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❖ Management

- Intravenous antibiotic therapy,
- Fluid replacement,
- Vestibular suppressant (meclizine)
- Antiemetic medications.
- Symptomatic for the viral one.

Anatomic and Physiologic overview of Nose surface

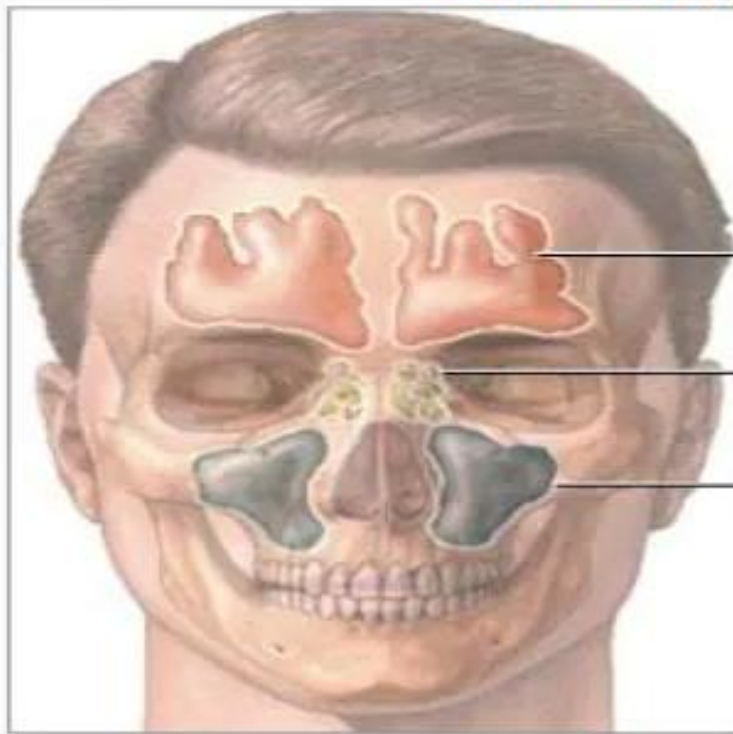
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Frontal sinus

Ethmoid sinuses

Maxillary sinus

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Disorder of Nose

1. Epistaxis

- It is hemorrhage from the nose.
- It can be;

A. Anterior Bleed

- Kiesselbach's plexus vessels.
- Easy to locate and treatment.

B. Posterior Bleed

- Larger vessels.
- Severe bleeding.
- Harder to locate and treatment.

Epistaxis cont....

❖ Etiology

- Dry cracked mucosal membrane, Trauma, Picking
- Blunt contact, Forceful nose blowing, Sneezing
- HTN, Chronic infection (AFI), Substance abuse
- Arteriosclerosis, Liver disease, Chronic bleeding disorder, Leukemia, Hemophilia, Anticoagulant Rx

Management

➤ Anterior

❖ Simple first aid

- Apply pressure for 5-10 minutes.
- Apply ice packs to nose & forehead.
- Sitting position leaning forward.
- Discourage swallowing blood.
- Medications
 - Topical vasoconstrictors (Cocaine , Neo-Synephrine, Adrenaline)
- Nasal spray or on cotton swab held against bleeding site

Management cont....

- Chemical cauterization
 - Silver nitrate
 - Gelfoam
- Topical anesthetic (pre-packing)
 - Tetracaine
 - Lidocaine
 - Cocaine
- Nasal Packing –Anterior
 - Petroleum gauze.
 - 24-72 hours commonly.

Management cont....

❖ Nasal Packing -Posterior

- Pack both anterior & posterior for 2-5 days.
- Monitor for hypoxemia.
- Administer oxygen as ordered.
- Frequent oral hygiene.
- Administer narcotic analgesics as ordered.
- Monitor for complications.
 - Toxic shock syndrome, Otitis media, Sinusitis

Management cont....

❖ Endoscopic Surgery

- Cauterizing bleeding vessel.
- Ligation of internal maxillary artery.

Nasal Polyps

- It is a benign grapelike growth of mucous membrane.
- Form in areas of dependent mucous membrane.
- Usually bilateral.
- Stem-like base makes them moveable.
- It may enlarge and cause nasal obstruction.

Management cont....

Management

❖ Medication;

- Topical corticosteroid nasal spray.
- Low-dose oral corticosteroids.

Management cont....

❖ Surgery;

- Polypectomy under local anesthesia.
- Nasal packing to control bleeding
- Avoid blowing nose 24-48 hours post removal of packing.
- Avoid straining at stool, vigorous coughing, strenuous exercise.
- Monitor for bleeding (Frequent swallowing , Visible blood at back of throat)
- Laser surgery to remove polyps.
- May require multiple surgeries as polyps tend to recur.

Deviated Septum

- May result from trauma
- May be present from birth
- Causes nasal obstruction

❖ **Management**

- Relief of airway obstruction.
- Repair visible deformity.
- Reshaping of nose by manipulation of septal cartilage
- Septoplasty or submucous resection.
- Rhinoplasty or surgical reconstruction of the nose.

Cont....

- Post operatively;
 - Bilateral Nasal packing for 72 hours.
 - Temporary plastic splint for 3-5 days.
 - Swelling subsides within 10-14 days.
 - Normal sensation returns within several months.

Rhinitis

- It is an inflammation of the mucous membranes of the nose.
- It has different classification;
 - Based on duration,
 - a) Acute
 - b) Chronic
 - Based on cause,
 - a) Allergic rhinitis /hay fever /:due to allergy.
 - b) Non-allergic rhinitis: following URTI (Bacteria and Viral)

Acute Rhinitis

- It is also known as Coryza or common cold
- Affects almost every one at some time and most often in the winter, with additional high incidence in early fall and spring.

❖ Cause

- Common etiology is virus.
- Each virus may have multiple strains.
- For example, there are over 100 strains of rhinovirus, which accounts for 50% of all colds.

Cont....

- Rhinovirus
- Corona virus
- Adenovirus
- Influenza virus
- Parainfluenza virus
- Echovirus
- Coxsakiervirus
- Respiratory syncytial virus (RSV),

Cont....

- It is highly contagious because virus is shed for about 2 days before the symptoms appear and after 3 days of the symptom.
- Common cold spread by;
 - Droplet nuclei from sneezing.
 - Contaminated hand or fomites.
- Secondary invasion by bacteria may cause;
 - Pneumonia
 - Acute bronchitis
 - Sinusitis
 - Otitis media

Cont....

Clinical manifestation

- Sneezing
- Nasal discharge (runny nose)
- Nasal obstruction/Nasal congestion
- Head ache
- Chilliness/Fever
- Nasal itchiness
- Shyness/nervousness
- Sore throat
- Malaise

Cont....

Medical management

- Usually self –limiting and lasts for about 1 week.
- Goal of management;
 1. To relieve symptoms
 2. Inhibit spread of the infection
 3. Reduce risk of bacterial complication

Cont....

- Adequate fluid intake.
- Encouraging rest.
- Preventing chilling.
- Increasing intake of vitamin C.
- Using expectorants as needed.
- Warm salt-water gargles soothe the sore throat.
- Nonsteroidal anti-inflammatory agents (NSAIDs) such as aspirin or ibuprofen.
- Antihistamines (chlorpheniramine maleate ,
diphenhydramine (Benadryl))

Cont....

- Topical (nasal) decongestant (e.g. oxymetazoline maleate, phenylephrine, pseudoephedrine orally.
- Zinc lozenges may reduce the duration of cold symptoms if taken within the first 24 hours of onset.
- Antimicrobial agents (antibiotics) **should not be used**
 - They do not affect the virus or reduce the incidence of bacterial complications.

Cont....

Nursing Management;

- Perform hand hygiene often.
- Use disposable tissues.
- Avoid crowds during the flu season.
- Avoid individuals with colds or respiratory infections.
- Obtain influenza vaccination, if recommended (especially if elderly or diagnosed with a chronic illness)

Chronic rhinitis

- A chronic inflammation of the nasal mucosal membrane characterized by increased nasal mucus.

❖ Cause

- Repeated acute infection or allergy.
- Vasomotor rhinitis (an instability of the autonomic nervous system caused by stress, tension , or some endocrine disorder).
- Chronic irritation by nasal drug

Clinical manifestation

- No acute symptom.
- Nasal obstruction (stiffness).
- Pressure in the nose.
- Polyp formation .
- Vertigo.

Management

❖ Nursing interventions

- Instruct a pt with allergic rhinitis to avoid allergens and irritants i.e. dusts, fumes, odor, powder spray
- Proper use and administration of medication.
- Obtain additional rest.
- Drink at least 2 to 32 times fluid daily.
- Use nasal spray or nose drops.

Sinusitis

- It is an inflammation of the mucous membranes in the sinuses.
- Sinusitis can be;
 1. Acute bacterial.
 2. Sub acute.
 3. Chronic.

Acute Sinusitis

- The most common types of acute sinusitis are;
- Allergic- Usually seasonal.
- Viral.
- Acute bacterial (Streptococcus pneumonia, haemophilus influenza, beta hemolytic streptococcus, klebsiella pneumonia and various anaerobic organisms).

Clinical manifestation

- Slowly developing pressure over the involved sinus
- General malaise
- Fever
- Systemic symptoms i.e., achiness
- Stuffy nose
- Persistent cough
- Postnasal drip
- Headache
- Redness and itching of the eye
- Sign of tooth infection

Clinical manifestation cont....

- In acute frontal and maxillary sinusitis , pain usually does not appear until 1 to 2 hours after awakening.
- It increases for 3 to 4 hours and then becomes less severe in the afternoon and evening usually this is due to increased drainage as result of gravity from standing during the day.
- Blood-tinged discharge from the nose in the first 24 to 48 hours.
- The discharge rapidly becomes thick, green, and copious, blocking the nose.

Diagnosis

- Hx.
- P/E;
- Tenderness in the involved sinus,
- Hyperemic and edematous nasal mucosa, and
- The turbinate's are enlarged.
- X-ray examination
 - Clouded sinus and fluid level is visible.

Managements

- Aim is to relief a pain and shrinkage of the nasal mucosa
- Medication
 - Analgesics i.e. . Ibuprofen.
 - Oral decongestant-pseudoephedrine.
 - Antibiotics i.e., Amoxicillin for 10 days to 14 days
 - Acute frontal sinusitis with pain, tenderness, and edema of the frontal or sphenoid sinus require hospitalization b/c of risk of intracranial complication or Osteomyelitis
 - High doses of IV antibiotic
 - nasal decongestant or by spray is needed.

Chronic Bacterial Sinusitis

- Chronic bacterial sinusitis develops when irreversible mucosa damage occurs.
- Damage can result from recurrent attacks of acute sinusitis or from suppurative sinusitis either being untreated or inadequately treated during the acute or sub acute phase.
- Etiology
 - S.aureus
 - H. influenza
 - Anaerobes (Klebsiella)

Clinical manifestation

- Nasal congestion
- Thick, green purulent discharge, present for at least 3 months
- Fever
- Facial pain
- Light headness /does not have headache

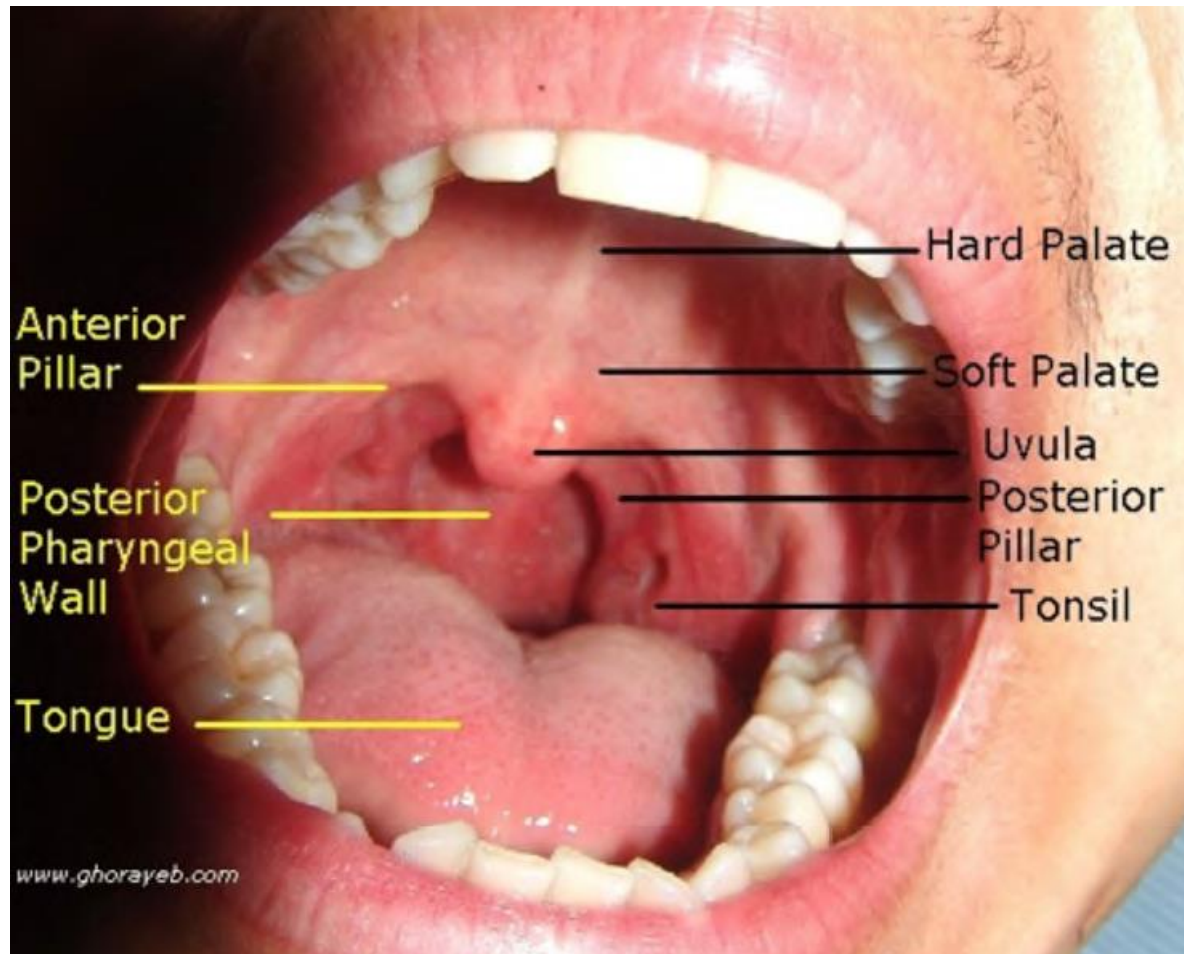
❖ **Diagnosis**

- Culture and sensitivity

Management

- Medication
 - Decongestant.
 - Antibiotic according to result of the culture.
 - Nasal saline irrigation and surgery are the major treatments.
- Pt. benefits from thing that increase the drainage.
 - Increasing the humidity (steam bath hot shower, facial sauna)
 - Increasing fluid intake applying local heat (hot wet packs).

Oral cavity



Pharangitis

❖ Acute pharyngitis

- Is a febrile inflammation of the throat
- Caused by 70% viral cause and 30% bacteria
 - i.e. hemolytic streptococci, staphylococci.
- It is the most common throat inflammation.
- A severe form of acute pharyngitis often is termed “Step throat” because of the frequency of streptococci as the causative organism.

Clinical manifestation

- Dryness of the throat
- Fiery red throat and pharyngeal membrane and tonsils
- Severe pain which leads to difficulty in swallowing
- Enlarged and tender cervical lymph nodes
- Fever, Malaise, Sore throat, Hoarseness, cough, Rhinitis

❖ **Diagnosis**

- Throat culture.
- Nasal swabbing and blood culture may be done.

Management

❖ Medical management

- Penicillin is a drug of choice.
- Erythromycin for 10 day.
- Liquid and soft diet.
- lozenges – to relieve local soreness .

❖ Nursing intervention

- Bed rest at febrile stage.
- Proper tissue disposal.

Management....

- Assess as for possible skin rash b/c pharyngitis may precede some other communicable disease.
- Warm saline gargles or irrigations are used.
- Analgesic medication.
- Prophylactic antibiotic therapy for pharyngitis in patients with a history of rheumatic fever or infective endocarditis to prevent re-infection.

Complication

- Sinusitis
- Otitis media
- Peritonsillar abscess
- Mastoiditis
- Cervical adenitis
- Rheumatic fever
- Rheumatic nephritis

Chronic pharyngitis

- It is a persistent inflammation of the pharynx.
- It is common in adults who work or live in dusty surroundings, use their voice to excess, suffer from chronic cough, and habitually use alcohol and tobacco.

Chronic pharyngitis....

- ❖ Three types of chronic pharyngitis are recognized:
 - Hypertrophic:-general thickening and congestion of the pharyngeal mucous membrane
 - Atrophic: probably a late stage of the first type (the membrane is thin, whitish, glistening, and at times wrinkled)
 - Chronic granular (“clergyman’s sore throat”): characterized by numerous swollen lymph follicles on the pharyngeal wall.

Clinical Manifestations

- A constant sense of irritation or fullness in the throat,
- Mucus that collects in the throat and can be expelled by coughing, and
- Difficulty swallowing.

Management

❖ **Medical Management;** Is based on

- Relieving symptoms,
 - Avoiding exposure to irritants, and
 - Correcting any upper respiratory, pulmonary, or cardiac condition that might be responsible for a chronic cough.
- Nasal sprays or medications containing ephedrine sulfate (Kondon's Nasal) or phenylephrine hydrochloride.
 - Antihistamine decongestant medications, such as Drixoral or Dimetapp, is taken orally every 4 to 6 hours.
 - Anti-inflammatory and analgesic agent like Aspirin or acetaminophen

Management.....

❖ **Nursing Management;**

- Avoid contact with others until the fever subsides.
- Alcohol, tobacco, second-hand smoke, and exposure to cold are avoided.
- The patient may minimize exposure to pollutants by wearing a disposable facemask.
- Drink plenty of fluids.
- Gargling with warm saline solutions
- Lozenges will keep the throat moistened.

Tonsillitis

❖ Tonsillitis

- Tonsillitis is inflammation and enlargement of the tonsil tissue.
- Tonsil tissue are situated on each side of the oropharynx

❖ Cause

- Could be viral (70%) or bacterial (30%)- As a result, antibiotics are typically not efficacious in the majority of cases.

Tonsillitis...

- Group A streptococcus is the most common organism associated with bacterial tonsillitis.
- Other bacterial examples include Haemophilus influenza, Streptococcus pneumonia and Staphylococci- the latter being more associated with dehydration and previous antibiotic use.

Symptoms

- Sore throat
- Odynophagia (painful swallowing) and dysphagia
- Earache (sometimes general muscle ache)
- Systemic upset: Malaise and headache
- Viral tonsillitis may present with milder symptoms

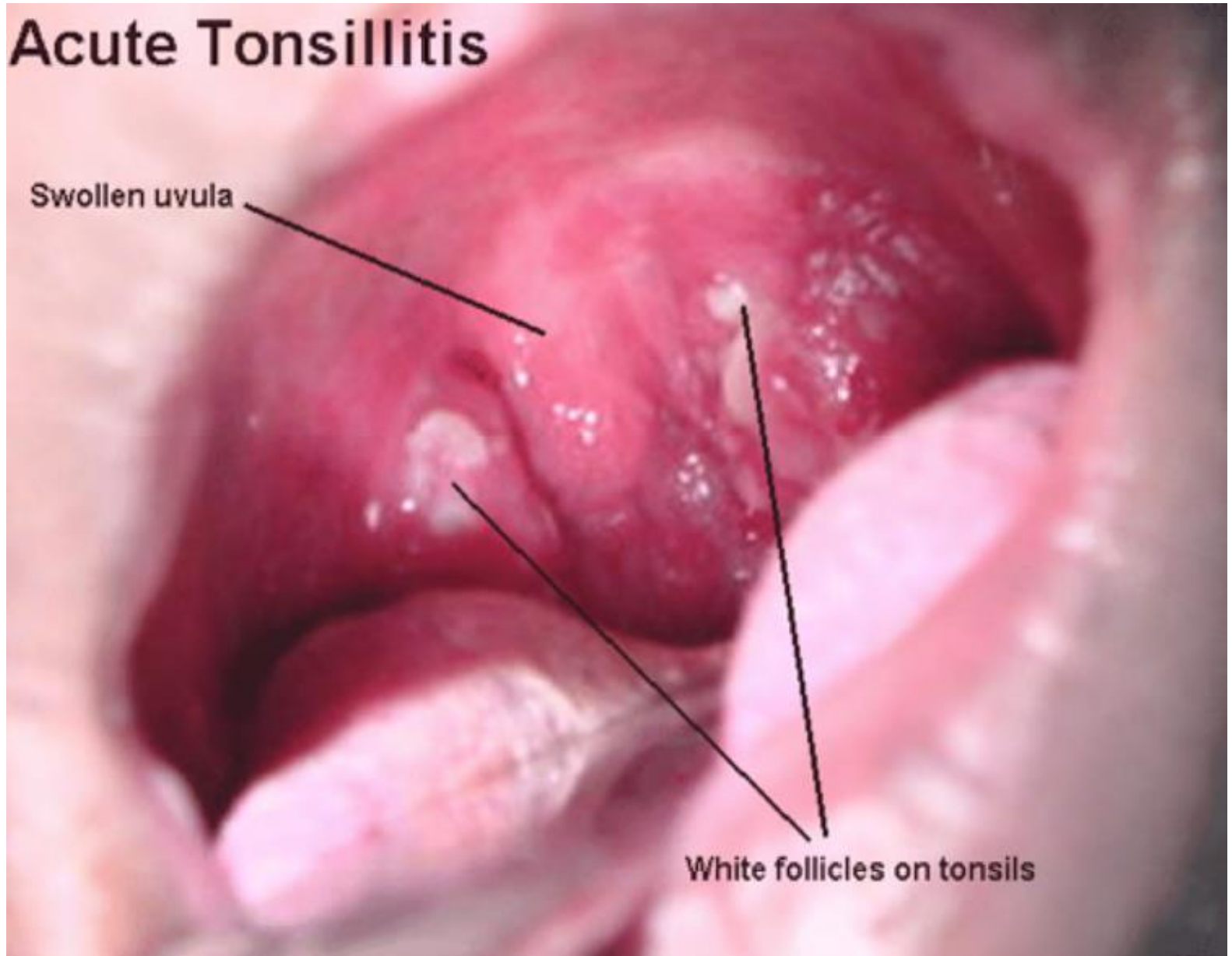
Signs

- Pyrexia/fever
- Swollen tonsils +/- exudate
- Thick or 'hot potato' voice with enlarged tonsils
- Presence of trismus (muscle spasm) indicates a peritonsillar abscess (quinsy)
- Bilateral cervical lymphadenopathy

Acute Tonsillitis

Swollen uvula

White follicles on tonsils



Left Sided Peritonsillar Abscess (Quinsy)



Natural history

- Resolves after 5-7 days, may recur after symptom free interval

❖ **Complications**

- Peritonsillar abscess
- Parapharyngeal and retropharyngeal abscesses- potentially life threatening complications of tonsillitis.
- They may present with non-resolving sore throat, systemic upset, neck stiffness and neck tenderness

DIAGNOSIS

- Hx.
- P/E.
- CBC, U&Es, CRP, Culture of tonsil swab.

Management

- There is a 50% chance of the tonsillitis being bacterial if:
 - Pus on tonsils (tonsillar exudate)
 - Pyrexia
 - No cough
 - Tender cervical lymph nodes
- Antibiotics should be given to those who meet 3 or 4 criteria.

- Penicillin V (phenoxymethypenicillin), 500mg QDS for 10 days.
- Penicillin G procaine -600,000- 1million IU IM for 10 dys
- Erythromycin is an alternative in penicillin allergic patients.
- Avoid amoxicillin- this causes a rash if the patient has glandular fever due to type IV hypersensitivity.

Treatment

- Mouth care may for comfort
- Tonsillectomy
- Indication
 - Repeated bout of tonsillitis. (3,5,7 for 3,2,1 year)
 - Respiratory obstruction.
 - Hypertrophy of the tonsils
 - Recurrent otitis media.
 - Peritonsillar abscess.

Adenoiditis

- Is inflammation of the adenoid tissue
- The adenoid consist of an abnormally large lymphoid tissue mass near the center of the posterior wall of the nasopharynx.
- Infection of the adenoids frequently accompanies acute tonsillitis.

❖ Clinical manifestation

- Mouth breathing, Earache, Draining ear, Yellowish exudates drain

Diagnosis

- Hx.
- P/E.
- Culture of tonsil swab.
- Audiometric examination (hearing loss).

Treatment

- Benzantine penicillin
- Mouth care may for comfort
- Adenoidectomy
- Indication
 - Respiratory obstruction.
 - Recurrent otitis media.
 - Peritonsillar abscess.

- Prophylaxis
- Prophylactically penicillin may be given.
- Educate on the continuation of the therapy.

Laryngitis

- It is inflammation of the larynx.
- Predisposing factor /associated to;
 - Voice abuse.
 - Exposure to dust.
 - Chemicals.
 - Smoke and other pollutants

Etiology

- Almost always is a virus
- Bacterial invasion may be
 - Acute rhinitis or Naso pharyngitis.
- The onset of infection may be associated with exposure to **sudden temperature change**.
- Diet as deficiencies
- Lack of immunity
- Laryngitis is common in the winter and is easily transmitted.

Clinical manifestation

- Chronic laryngitis
 - Persistent hoarseness.
 - Hoarseness or complete loss of voice (aphonia).
 - Severe may be a complication of chronic sinusitis and chronic bronchitis.

Management

- Resting the voice,
- Avoid smoking,
- Resting in bed , and
- Inhaling cool steam or an aerosol

❖ **For chronic laryngitis**

- Resting the voice.
- Eliminating any primary respiratory tract infection
- Restricting smoking.

❖ Nursing interventions

- The patient is instructed to rest the voice and to maintain a well humidified environment.
- High fluid intake.